

Introduction

Modern data centers require highly scalable and low-latency network architectures to support rapidly growing cloud workloads.

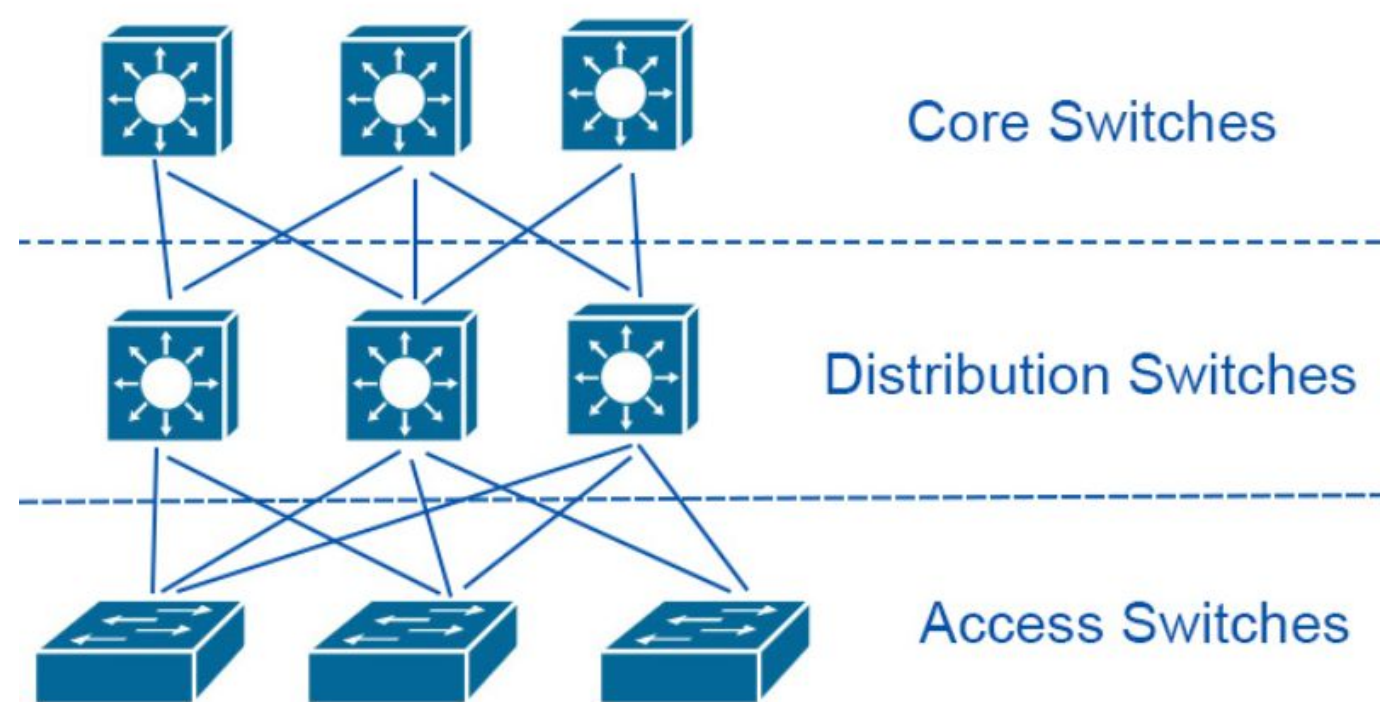


Figure 1. Core-Distribution-Access Architecture

- Traditional Three-tier Data Center Architecture networks were designed mainly for north–south traffic.
- The rise of virtualization, distributed applications, and microservices has significantly increased east–west traffic between servers inside the data center.
- Hierarchical architectures (Core / Distribution / Access) can introduce network bottlenecks and variable latency, limiting performance and scalability.
- The Leaf–Spine Architecture was introduced to address these challenges by providing high bandwidth and horizontal scalability.

Leaf–Spine Architecture Principle

The Leaf–Spine Architecture organizes the data center network into two switching layers to provide predictable latency and high bandwidth between any pair of servers.

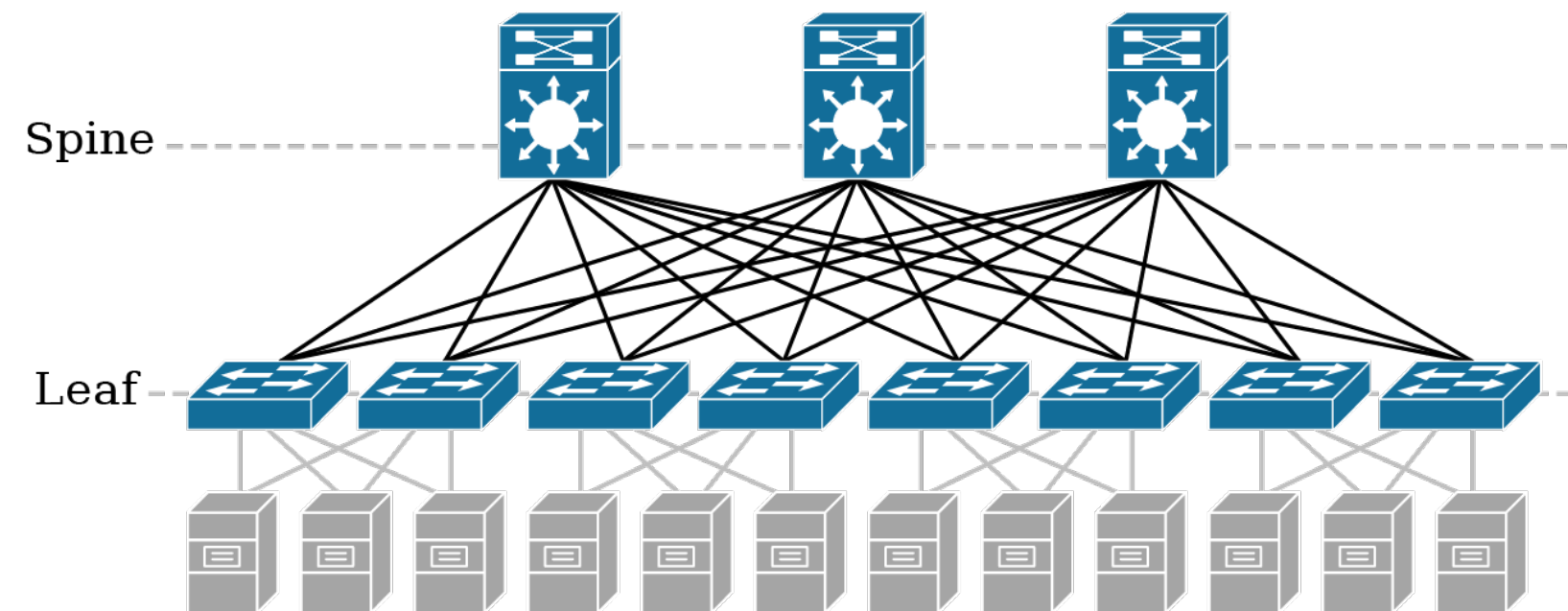


Figure 2. Leaf-Spine Architecture

Leaf Layer:

- Consists of Top-of-Rack (ToR) switches connecting servers within each rack.
- Each leaf switch provides access connectivity for compute and storage nodes.
- Every leaf switch connects to all spine switches, ensuring multiple available paths.

Spine layer:

- Acts as the high-speed backbone of the data center network.
- Spine switches interconnect all leaf switches but do not connect directly to servers.
- Provides high-capacity switching to forward traffic between different racks.

Underlay-Overlay Model

In this architecture, the physical connectivity is separated from virtual network services using an underlay–overlay model.

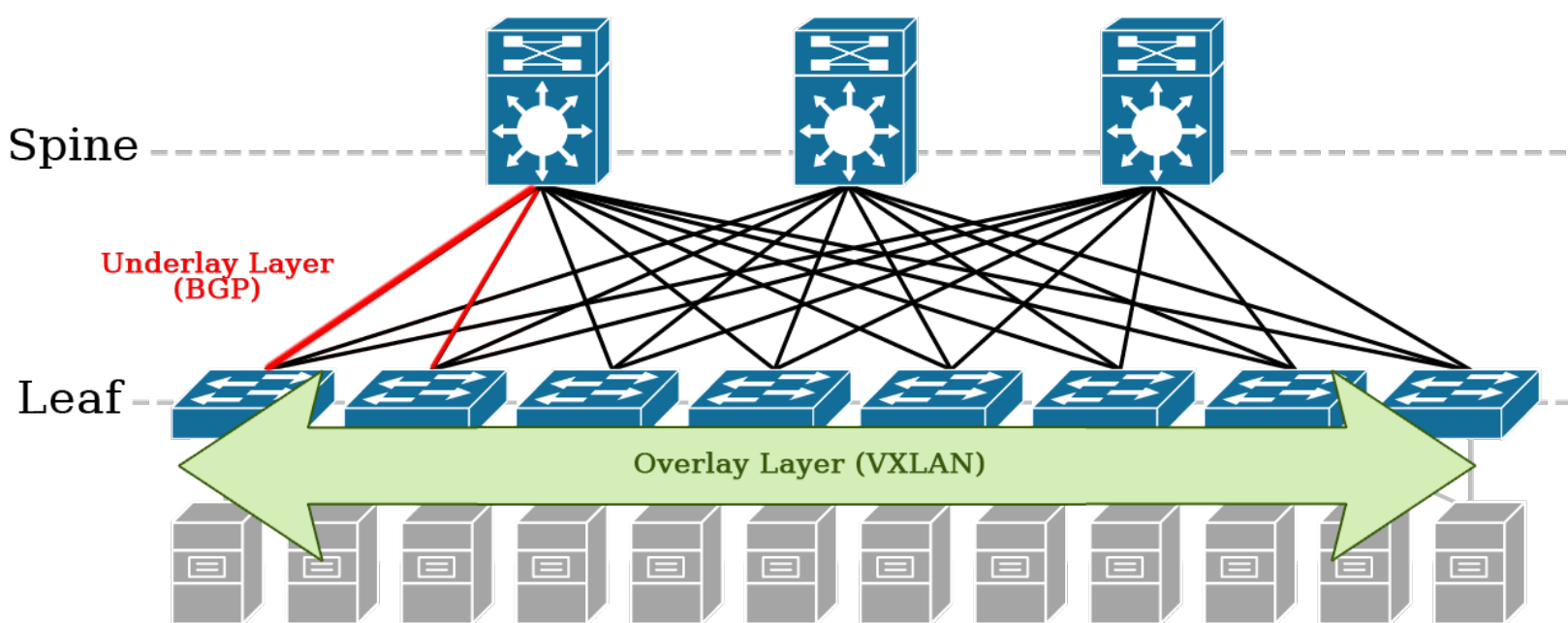


Figure 3. Underlay-Overlay Model

Underlay Layer:

- The underlay represents the physical IP fabric of the data center built on the Leaf–Spine Architecture.
- It consists of leaf and spine switches, physical links, and routing protocols (OSPF, IS-IS, BGP) that ensure IP reachability between all switches.
- The main objective of the underlay is to provide a highly available, low-latency transport network across the fabric.

Overlay layer:

- The overlay is a logical network built on top of the underlay infrastructure.
- It enables virtualized Layer-2 or Layer-3 connectivity independent of the physical topology.
- Overlay networks typically use encapsulation mechanisms such as VXLAN to transport traffic across the IP fabric.

Underlay Design: BGP-Based IP Network

BGP is widely adopted in modern data center fabrics due to its scalability, flexibility, and operational efficiency. Unlike OSPF and IS-IS, BGP avoids large flooding domains and relies on incremental route updates.

When deploying BGP, we select between Internal BGP (IBGP) and External BGP (EBGP). The biggest difference between IBGP and EBGP lies in the way they use ASs

iBGP Underlay Model:

In iBGP, all leaf and spine switches operate within the same AS. Leaf and spine switches establish iBGP peerings to exchange routing information.

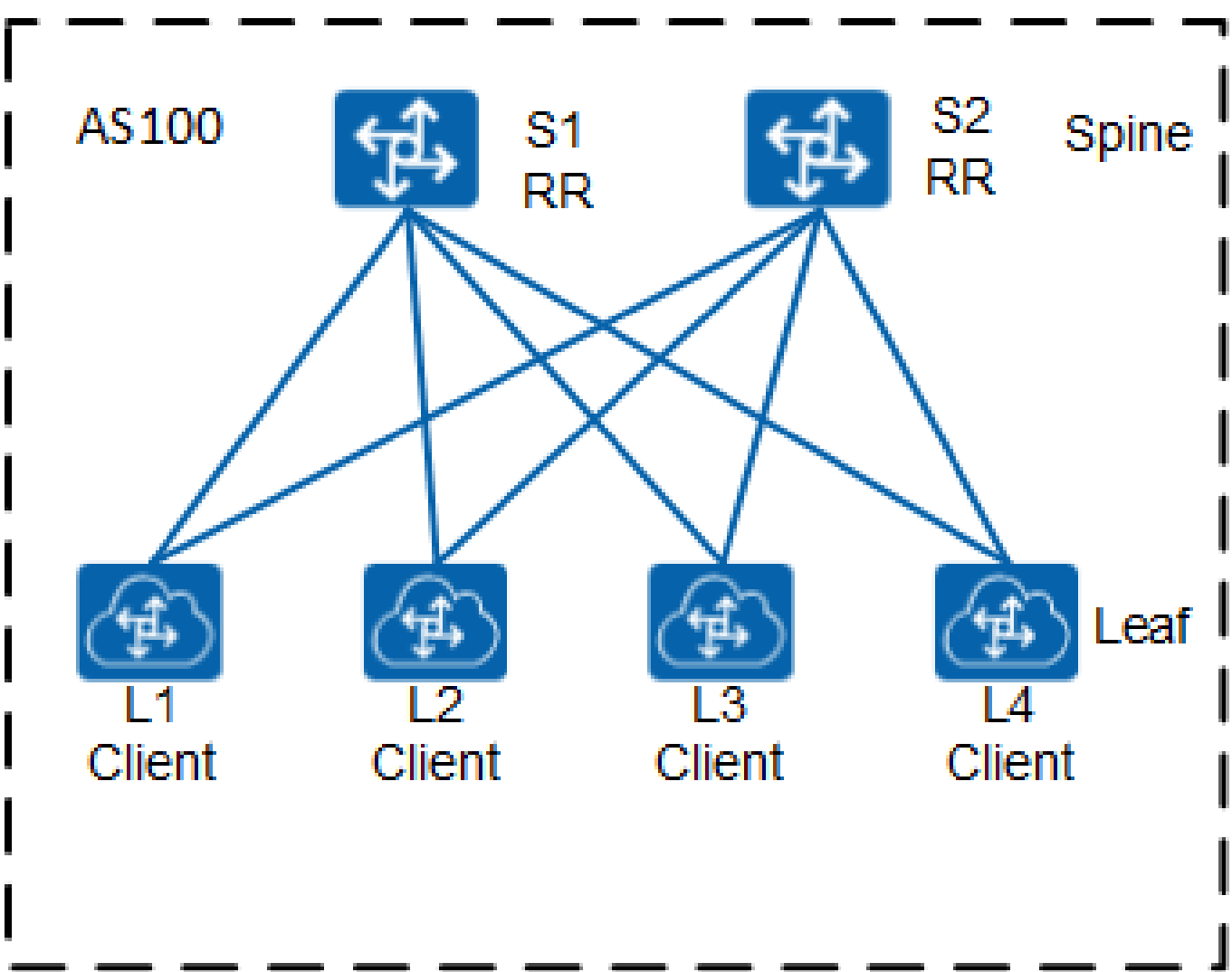


Figure 4. iBGP Underlay Model

Split-Horizon Rule:

- iBGP prevents a router from advertising a route learned from one iBGP peer to another iBGP peer.
- Route reflectors solve this limitation by reflecting routes from one leaf to other leaves.

Next-Hop Reachability:

- iBGP does not modify the next-hop attribute when propagating routes, it remains the originating leaf loopback address.
- To ensure reachability, each switch advertises its loopback address in BGP, allowing recursive resolution and load balancing using Equal-Cost Multi-Path Routing (ECMP).

eBGP Underlay Model:

In eBGP, each switch typically belongs to an Autonomous System (AS). Routing loops are prevented via the **AS_PATH** attribute, which records the ASes a route traverses. A router rejects any route containing its own AS number in the AS_PATH list.

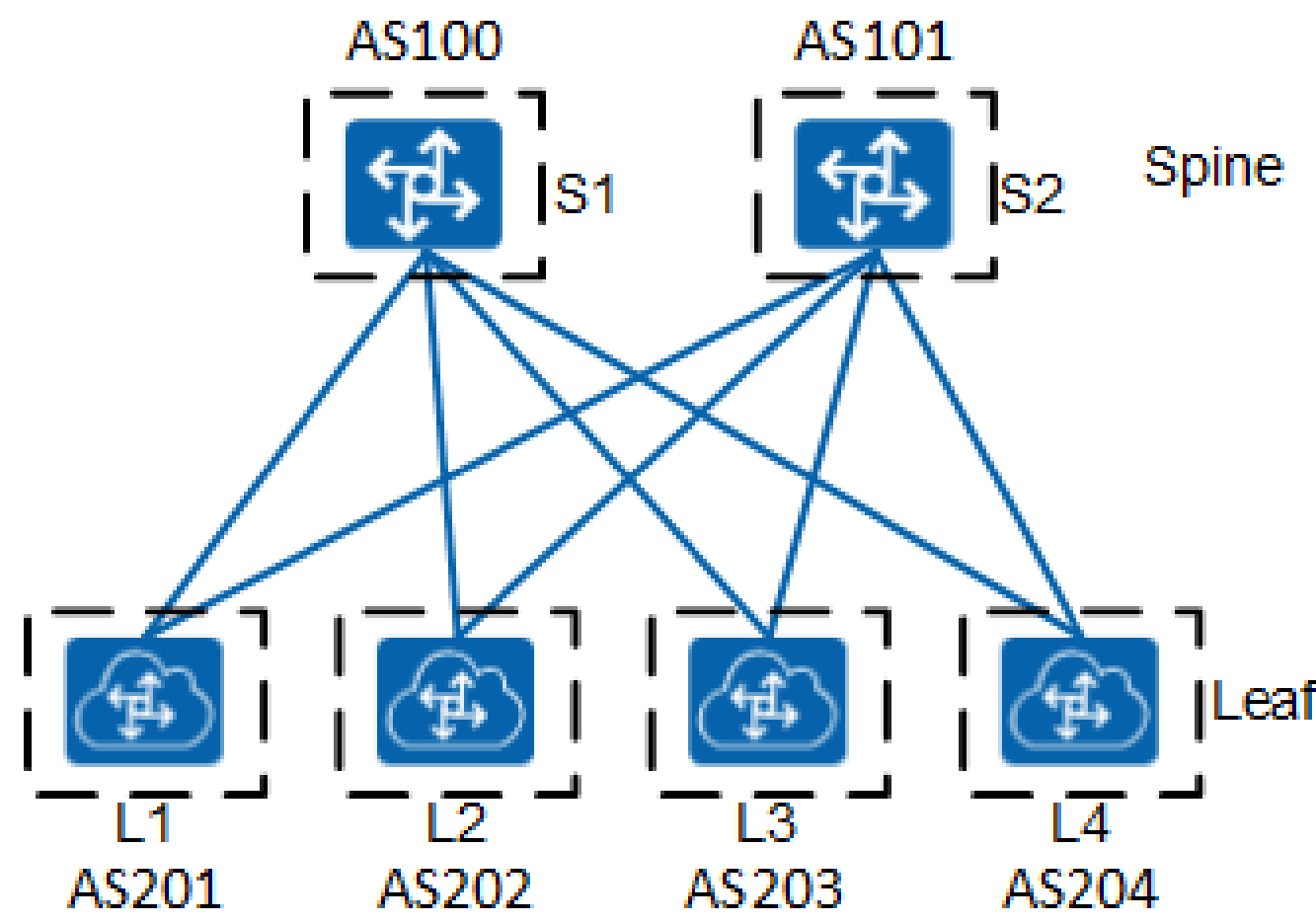


Figure 5. eBGP Underlay Model

eBGP Design Alternatives:

Different AS allocation schemes can be used depending on network design and scale, such as sharing one AS for spines and assigning unique AS numbers to leaves. BGP loop prevention works via AS_PATH, and 2-byte private AS numbers (64512–65535) are typically used.

Equal-Cost Multi-Path (ECMP)

ECMP enables multiple equal-cost paths simultaneously, fully utilizing the leaf–spine fabric bandwidth. Unlike traditional networks that rely on Spanning Tree Protocol (STP) to block redundant links, ECMP allows all paths to be used for forwarding.

Load Balancing (5-tuple):

Traffic is distributed across multiple paths using a hash computed from the packet **5-tuple**:

- Source IP
- Destination IP
- Source Port
- Destination Port
- Protocol

Benefits:

By leveraging all available paths, ECMP provides several advantages:

- High bandwidth utilization
- Fast convergence after failures
- Linear scalability by adding new spine switches

VXLAN Overlay

VXLAN encapsulates complete Ethernet frames inside UDP/IP packets, allowing L2 segments to span an L3 routed fabric. The underlay forwards encapsulated traffic as ordinary IP flows, the virtual segment is unaware of the physical hops between VTEPs.

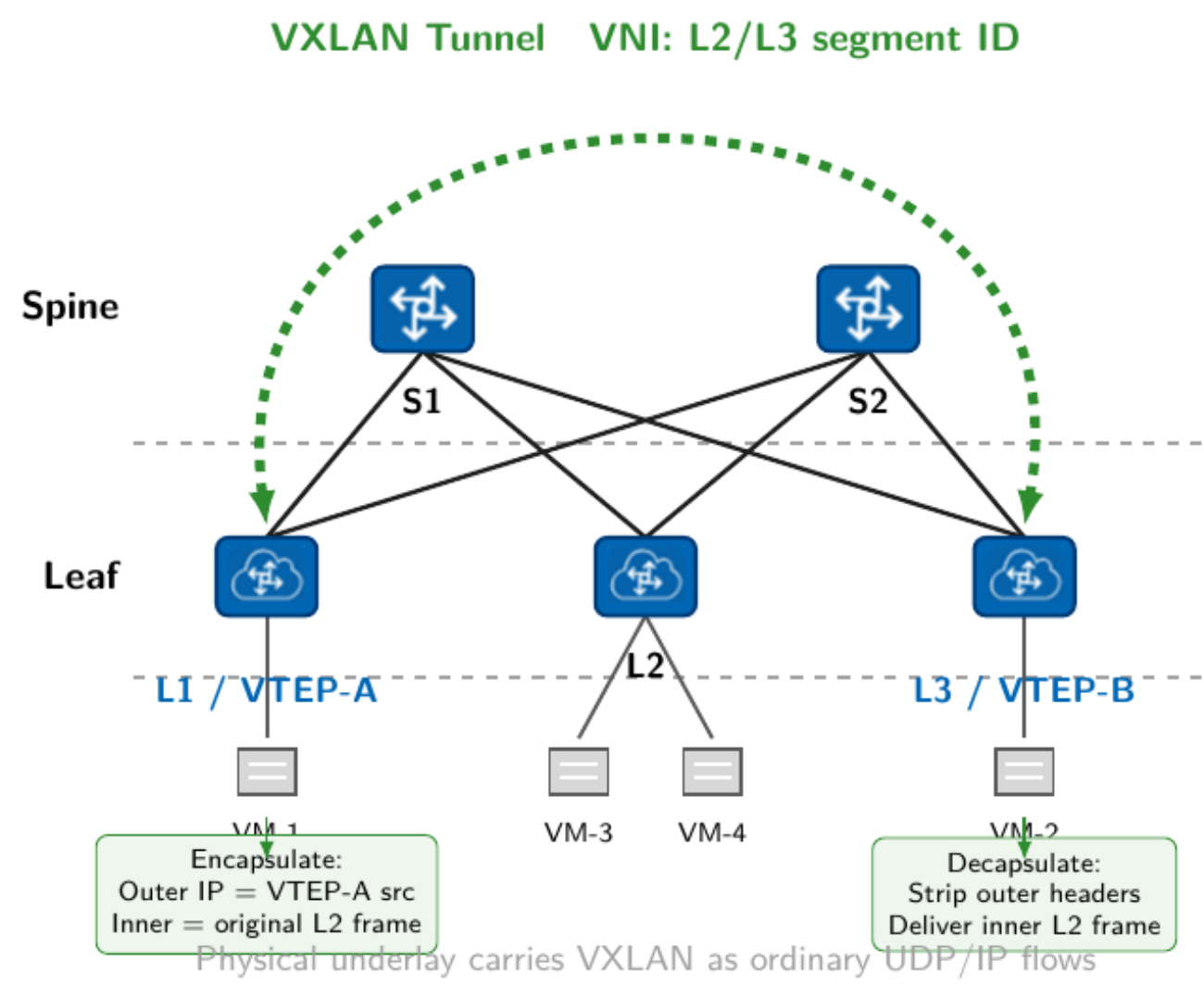


Figure 6. VXLAN tunnel between VTEPs over the underlay fabric

VXLAN (Data Plane):

- **VXLAN Tunnel Endpoints (VTEP):** Leaf switches act as Virtual Tunnel Endpoints (VTEPs), handling MAC-in-UDP encapsulation and decapsulation (destination port 4789). Spine switches simply route IP packets and remain unaware of the virtual traffic.
 - **VXLAN Network Identifier (VNI):** A 24-bit field in the VXLAN header, enabling over 16 million isolated virtual networks, far beyond the 4,096 limit of traditional VLANs.
- ### Multi-tenancy:
- **Logical Isolation:** Tenants are separated using VRFs (Virtual Routing and Forwarding) for Layer 3 isolation and VNIs for Layer 2 segmentation.
 - **Symmetric Bridging & Routing:** Local VLANs map to unique L2 VNIs for isolated bridging, while routed traffic is handled inside per-tenant VRFs using L3 VNIs, ensuring end-to-end isolation even when tenants use identical network configurations.

EVPN Control Plane

EVPN replaces probabilistic, traffic-based MAC learning with deterministic, BGP-distributed forwarding state, every switch knows endpoint locations *before* any data traffic is sent.

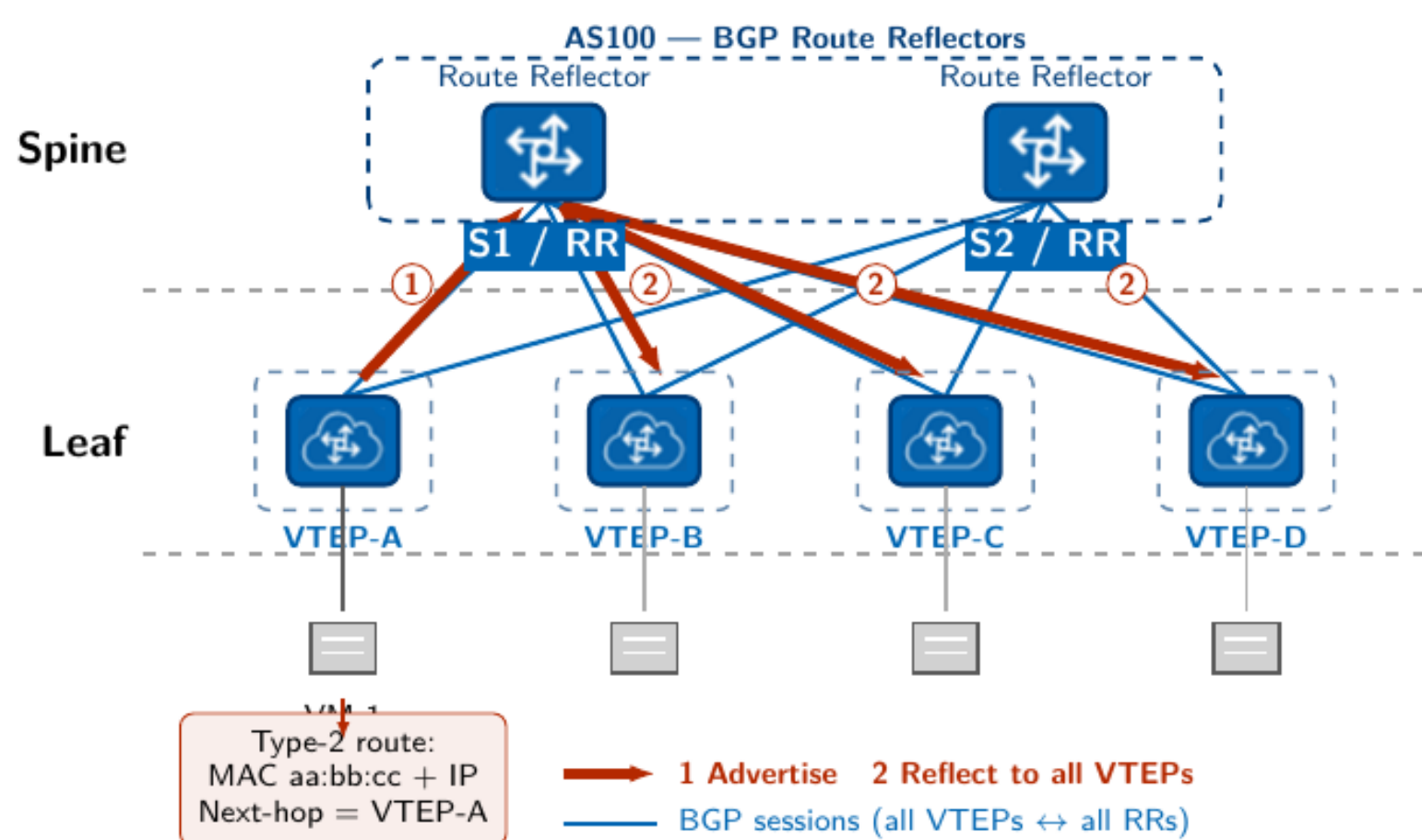


Figure 7. EVPN Type-2 MAC/IP distribution via BGP route reflectors

Distributed learning via BGP:

EVPN distributes endpoint locations (Type-2), BUM group membership (Type-3), and external routes (Type-5). All via the same BGP sessions already used by the underlay.

Key Mechanisms:

- **ARP suppression:** requests answered locally by the ingress leaf, the dominant broadcast source is eliminated at scale.
- **VM mobility:** the new leaf advertises an updated binding; all switches converge automatically, no ARP storms, no manual intervention.

Scalability

The Leaf-Spine fabric scales incrementally without global reconfiguration, growth is bounded by hardware capacity, not by protocol design.

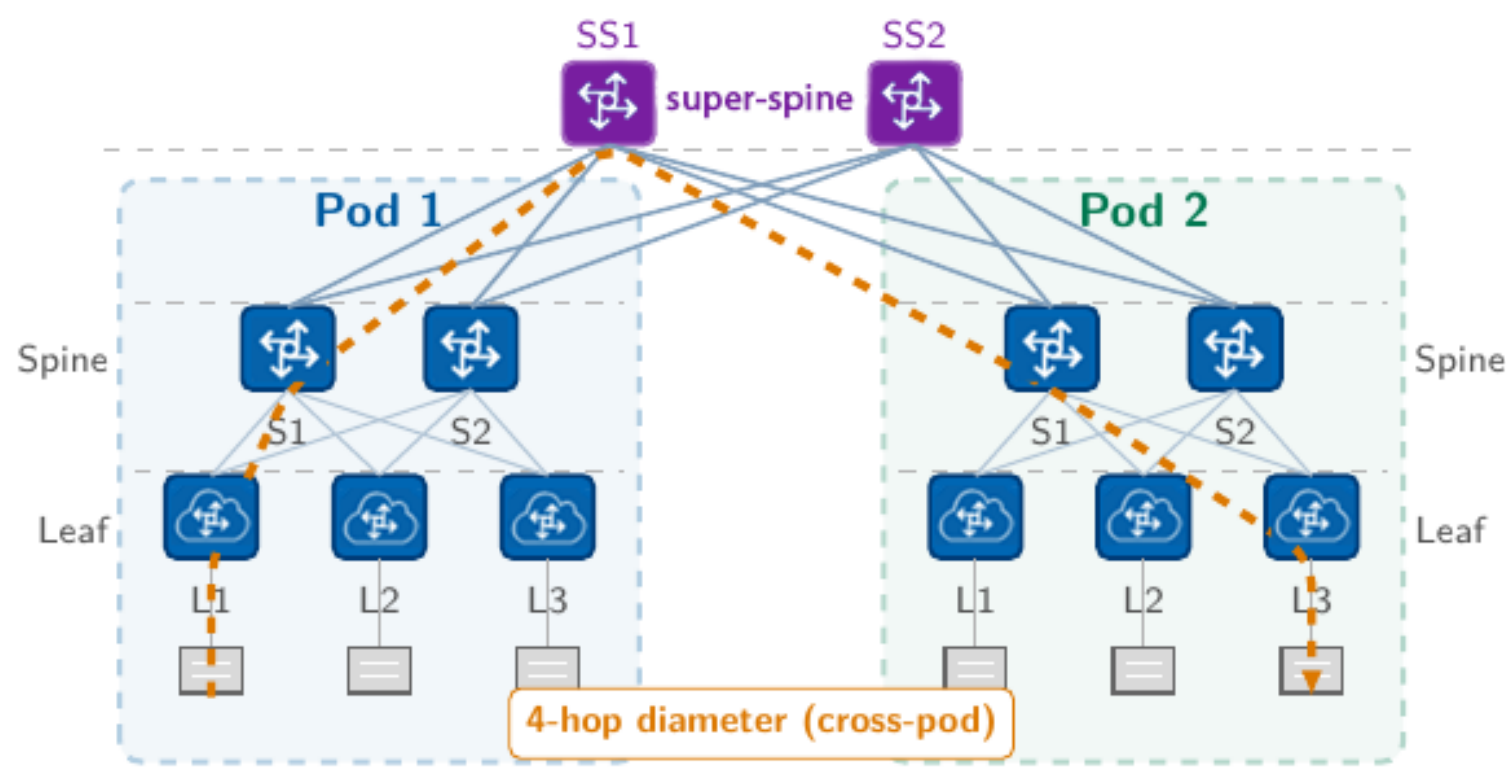


Figure 8. Super-spine layer connecting two pods (4-hop cross-pod path)

Pod Scaling:

- Add leaves for more hosts; add spines for more bandwidth, no reconfiguration of existing devices.
- Every leaf is equidistant from every spine: the bipartite structure is preserved at any scale.

Multi-Pod Growth:

- A super-spine layer connects pods when spine port count is exhausted; diameter grows from 2 to 4 hops.
- Ceiling: TCAM capacity on leaves and BGP processing capacity on spines, not protocol limits.

Advantages & Disadvantages

Advantages:

- **Predictable Performance at Scale:** The uniform two-hop design guarantees consistent latency and allows seamless horizontal scaling without network reconfiguration.
- **Simplified, Any-to-Any Multi-Tenancy:** BGP and VXLAN unify routing and securely isolate tenants across the fabric, replacing complex legacy protocols.

Disadvantages:

- **Strict Hardware Dependencies:** The architecture requires specific, high-end switches with adequate TCAM, VXLAN offloading, and port density to function properly.
- **High Initial Complexity:** Designing and deploying a BGP EVPN/VXLAN fabric involves a steep learning curve and demands advanced networking expertise.

Vendor Comparison

Implementing a Leaf–Spine Architecture requires high-performance switches. These switches must support hardware VXLAN. They need **high port density** to connect multiple racks and spines(32 to 128 per switch). **High-bandwidth** interfaces (25, 40, 100, or 400 GbE) are needed to sustain east–west traffic. Switches should also support modern protocols like EVPN and BGP.

Vendor	OS	Automation / Management	Strengths / Considerations
Cisco	NX-OS	ACI / DCNM	Large ecosystem, strong enterprise support; High cost, complex licensing, proprietary (ACI)
Arista	EOS	CloudVision / AVD	Very stable OS, strong automation; Smaller global footprint than Cisco
Juniper	Junos	Apstra	Structured OS, strong automation; Industry uncertainty due to HPE acquisition
Huawei	VRP / CloudEngine OS	iMaster NCE / eSight	Competitive pricing, strong data center, good automation tools; Geopolitical restrictions, limited adoption and complex licensing

Table 1. Vendor comparison for data center networking.

Conclusion

The Leaf-Spine architecture with VXLAN and EVPN overcomes traditional three-tier limitations by delivering predictable two-hop paths, horizontal scalability, and multi-tenant isolation. Its unified BGP control plane simplifies routing, inherently enabling any-to-any connectivity without legacy complexity. Ultimately, selecting hardware with adequate TCAM, VXLAN offload, and port density is crucial to ensure the fabric scales seamlessly alongside the workloads.

References

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- [4] Ma Tengfeng, Zhu Hualian, Li Shuangshuang, Zong Yue, Zhou Tingting, and Gao Yangyang. *Huawei DCN Design Guide*. Huawei Technologies Co., Ltd., Shenzhen, China, issue 02 edition, 2025.